	ISO 26262 CookBook	Issue/Amendment 5.8	Page 1/53
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1 Objective

- Guiding projects through different process steps required for ISO26262 implementation in PS-SC Process Library.
- Mapping of work products required for ISO26262 compliance to templates, artifacts and processes in PS-SC Process Library.
- Hints for specifying difference between ASIL B/C/D requirements.

Note – This cookbook is not an alternative to PS-SC Process Library. Please refer to PS-SC Process Library for latest updates in the processes.

2 Scope

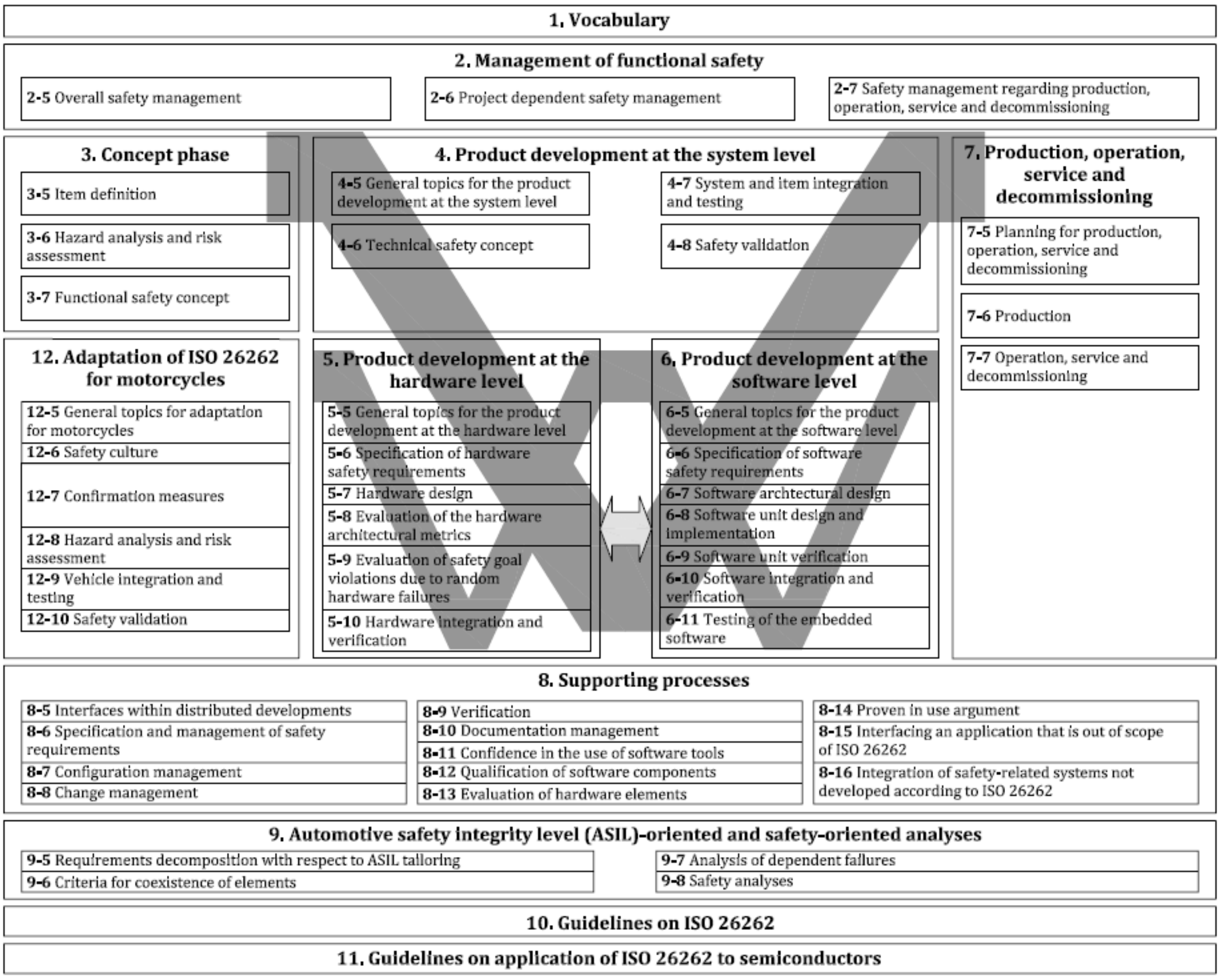
PS-SC ECU/VCU development projects based on PS-SC Process Library V7.0 and later (stages V6 based).

3 Overview of ISO 26262

ISO26262 is intended to be applied to safety-related systems that include one or more electrical and/or electronic (E/E) systems installed in series production road vehicles.

ISO26262: 2018 consists of the following parts

- Part 1: Vocabulary
- Part 2: Management of functional safety
- Part 3: Concept phase
- Part 4: Product development at the system level
- Part 5: Product development at the hardware level
- Part 6: Product development at the software level
- Part 7: Production, operation, service and decommissioning
- Part 8: Supporting processes
- Part 9: Automotive Safety Integrity Level (ASIL)-oriented and safety-oriented analyses
- Part 10: Guidelines on ISO 26262
- Part 11: Guidelines on application of ISO 26262 to semiconductors
- Part 12. Adaptation of ISO 26262 for motorcycles.



(Ref: ISO 26262-2:2018 –Overview of the ISO 26262 series of standards)

Please visit Web Based Training (WBT) on ISO26262 for more details on each of the part.

WBT (English): in TrainM, see **[SP-VS-ISO26262-WBT] ISO 26262 Introduction-B**

[SP-VS-ISO26262-WBT] ISO 26262 Introduction-B
(WBT 16758)

CONTENT:- Introduction in Functional Safety of Road Vehicles- Overview over the normative requirements based on ISO 26262:2018 (2nd edition) Target level for measure: BasicCompetence: Technology

4 ASIL Decomposition Strategy @ PS-SC and Safety Goal

Monitoring functions are implemented in the control unit as per the 3-level safety concept. ASIL decomposition is applied to these 3 levels as given below. These are Bosch preferred decompositions.

ASIL B Decomposition → Level 1 @ QM
Level 2, 3 @ ASIL B

ASIL C/D Decomposition → Level 1 @ QM
Level 2, 3 @ ASIL C

Safety Goal → Control of unintended acceleration and vehicle movement

Note – Projects may have different safety goals and ASIL decomposition strategy, based on customer requirements. This should be discussed and agreed with monitoring platform development team (PS-SC/ESC department).

5 PS-SC Process Library and ISO 26262

This section describes how requirements from different parts of ISO 26262 standard are realized in PS-SC Process Library and mapping of ISO 26262 work products to artefacts and processes.

Part 1 – Vocabulary, Part 10 – Guidelines, Part 11- Guidelines on application of ISO 26262 to semiconductors and Part 12- Adaptation of ISO 26262 for motorcycles are not covered in this cookbook.

Part 7 - Not covered in this cookbook as the production is not in PS-SC responsibility.

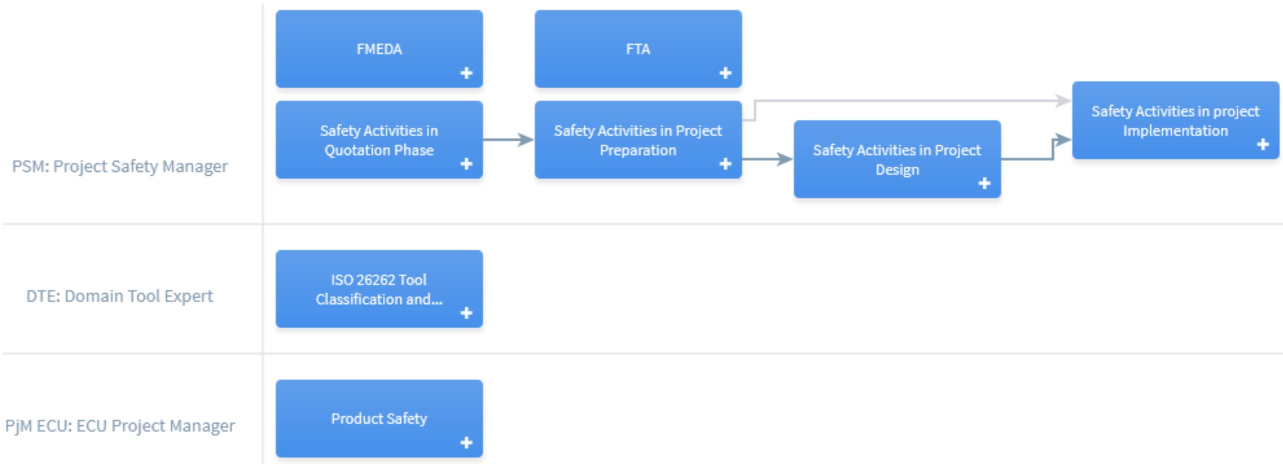
Part 2 to 6 are the core development processes.

Part 8 & 9 are supporting processes including safety analysis and are covered along with core development processes in this cookbook.

Functional Safety 'View' in process library.

The Functional Safety view represents the overall structure of the ISO 26262 series of standards, with a mapping to the individual process areas in the PLIB. Click on title of the process group to be directed to the corresponding process page.

Process Group: Functional Safety



Also, the Role based representation and Project phase wise representation explains the mapping of the parts/clauses of the ISO 26262 standard to the PS-SC product development lifecycle.

Note: PTSA 2.x Proclib details are provided. For PTSA1.x and Cbd Projects, refer the corresponding Proclib.

Part 2 & 8: Management of Functional Safety & Supporting Processes

This part describes the requirements for the organizations that are responsible for the safety lifecycle, or that perform safety activities in the safety lifecycle.

Safety Plan – It is one of the main work products of Part 2. Safety Plan contains relevant work products that are required for ISO 26262 compliance. Planning of these work products are made in safety plan against each quality milestone (QGCx/QGPx) of the project.

Note: A RASIC chart is available (as a good practice document) which lists out important activities for safety plan and the respective roles e.g. PSM, SWSD-Mo, CalEng Mo, PRes-Mo etc.

Link: good_practise_safety_roles_RASIC.xlsx

The role PSM is equivalent to the 'Safety Manager' role defined in ISO 26262 standard.

There is no 'Safety Engineer' role as such defined at PS-SC. Roles listed in the above RASIC e.g. SWSD-Mo, FND-Mo, FA-Mo, P-Res Mo, PjM HW etc. executes the activities of a safety engineer/safety expert at various stage of product development

Safety Case – It is the collection of safety relevant work products planned in safety plan. Safety case release is required before the project release QGF/QGP2. It is done with Safety case review with required level of independency as expected by ISO 26262 standard. Safety case review protocol is filled during the safety case confirmation review and is an evidence of the safety case release for the project.

A summary of the safety case is also documented in the Safety Plan and considered as safety argument

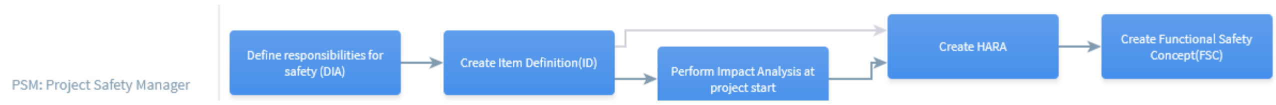
Impact Analysis - An initial Impact Analysis is performed in the planning phase, based on the modifications of the item considered at the start of the project. Design modifications during the development are implemented through a change management process. However, it is necessary to update the impact analysis sheet to document the impact on functional safety.

Tools Classification and Qualification – A software tool used in the development of safety relevant functions (system or software or hardware) including verification & validation activities, shall be classified and if required, qualified. The responsibility lies with DTE (Domain Tool Expert) or also known as 'tool responsible' in the department/project.

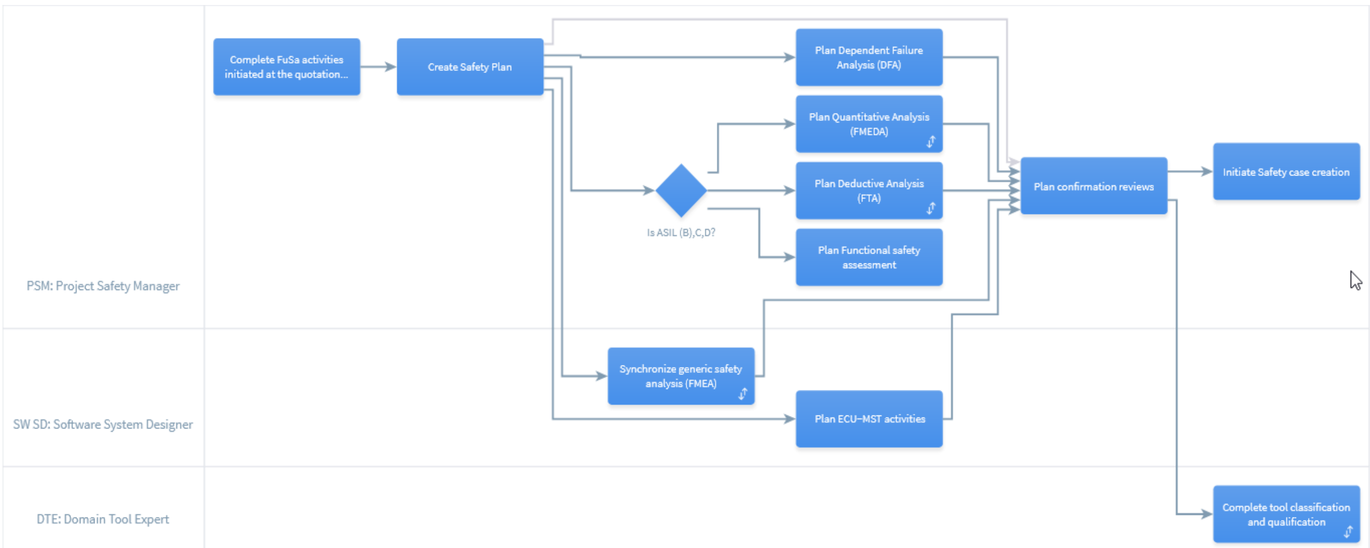
Hint: For VCU Integration platform where the scope is to deliver BSW and HW alone, limited tests can be done against functional requirements for HW monitoring using ECU-MST. For other use cases, tool detection methods for safety related errors are to be identified for the supported use cases. If tool detection is not possible, the same needs to be highlighted in the classification document and consequences/risks are to be discussed with OEM/relevant stakeholders.

In PS-SC Process Library, Safety related activities is specified in the **Process -> Workflows and Activities -> Functional Safety (FuSa)**

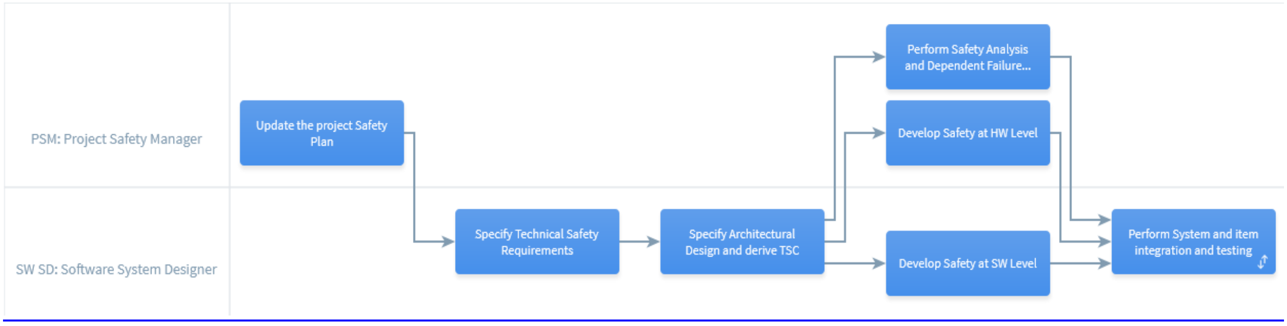
Safety Activities in Quotation Phase



Safety Activities in Project Preparation



Safety Activities in Project Design



Safety Activities in project Implementation

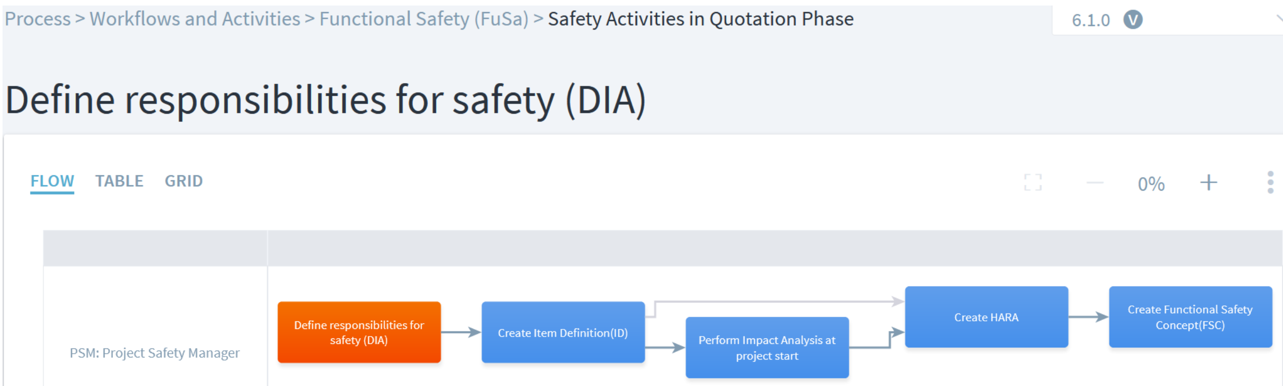
Development Interface Agreement (DIA) – Must be executed between customer and supplier. It is an agreement between customer and supplier in which the responsibilities for ISO 26262 activities, evidence or work products to be exchanged by each party are specified. If customer does not demand ISO 26262 compliance, DIA need not be executed. However written confirmation on the same shall be obtained from customer.

Recommendation: document the activities in Bosch scope using DIA template from process library and share with customer. This could be useful later in the project, for example to reference in TCD.

Hint: Additional attachments to DIA is possible at project level. For example, detailed monitoring split, interface and dependencies between L1 & L2 SW etc. can be discussed and agreed with customer during DIA preparation.

DIA is specified in **Process -> Workflows and Activities -> Functional Safety (FuSa) -> Safety Activities in Quotation Phase**

Define responsibilities for safety (DIA)



5.1.1 Mapping of work products from Part 2 and 8 to Process Library

ISO section	Work product	Remarks	Responsible	Links/Examples
Overall safety management				
2-5.5.1	Organization specific rules and processes for functional safety	Central directives, work instruction & Processes for functional safety	PS/QMM, PT Safety Team, PS-SC/EPM-EC	RB - CDQ 0214 PLIB QMS-Powertrain
2-5.5.2	Evidence of competence management	Role based Competence Plan Further link to BUM decision on ISO26262 WBT and link to management of competence can be given (access only to GrL) WBT on ISO26262 and certificate of successful completion by associates	Group Leader	Role Based Competency Plan
2-5.5.3	Evidence of quality management	RB fulfils the requirement from the IATF 16949, ISO 9001. This is checked by CDQ and is confirmed with a certificate, available with PS/QMM.	PS-SC/QMM	LINK

2-5.5.4	Identified safety anomaly reports	eMFace Report as part of ECU MST (eMFace Report should list all identified anomalies (e.g. buggy versions of FC and explanation why the usage is allowed))	PSM	eMface analysis report
Safety management during item development				
2-6.5.1	Impact Analysis at the item level	Carried out at the beginning of the project to check which parts and functions are affected. Further Impact Analysis in the project development stages are covered in RQ1. Safety relevant requirements should be added to the Impact Analysis sheet.	PjM ECU/ PSM	Impact Analysis
2-6.5.2	Impact Analysis at element level	For PS-SC; ECU is Item and element -> no separate Impact Analysis is required	NA	NA
2-6.5.3	Safety plan	Planning of safety related activities, work products and procedures with dependencies, starting point, duration, target dates, responsible in order to achieve Functional Safety. Process and template available in PLIB	PSM	SafetyPlan
2-6.5.4	Safety Case + Confirmation measure reports	Safety case is compilation of all work products in Safety Plan till series release. Safety case confirmation review is done to check the completeness of safety case and decide on the release.	PSM	SafetyPlan SafetyCase Safety case confirmation review

2-6.5.5	Confirmation measure reports	Confirmation reports from various confirmation reviews specified in safety plan. See overview at the end of the table.	PSM	
2-6.5.6	Release for production report	Released safety case, confirmation reviews, safety relevant tests in PVP, functional safety assessment report (in case of ASIL C, D), successfully passed Quality Gate.	PSM	
2-7.5.1	Evidence of field monitoring	PS/QMC tracks this as per CDQ0214 and in case of failure then the CDQ0904 is applied	NA	RB - CDQ 0214
8-5.5.2	Development Interface Agreement (DIA)	To be created between customer and supplier. Process and Template available in PLIB	PSM	DIA
Configuration management				
8-7.5.1	Configuration management plan	Configuration management processes and CM plan in PLIB	PjM ECU	CM Process Project specific CM-Plan
Change management				
8-8.5.1 8-8.5.3 8-8.5.2	Change management plan Change request Impact analysis and the change request plan	Done as per Req Management process, specified in PLIB and CDQ0404. Process for series changes in HW needs to be adopted for safety related components, in discussion.	PjM ECU	Change request management

8-8.5.4	Change report			
Documentation				
8-10.5.1	Document management plan	Specified in project CM plan. Further all the work products are specified in safety plan are updated at each milestone and are archived for 35 years as a part of safety case. Usage of ILM/ sharepoint is the recommended method in PS-SC.	PjM ECU	CM Plan Archiving, Retrieval
8-10.5.2	Documentation guideline requirements	Specified in the CM processes, work instructions and in CD-02981	PjM ECU	CM Process CD-02981
8-11.5.1	Software tool criteria evaluation report	Tool List Generic tools' database Tool classification report	DTE (Domain Tool Expert)	Tool List ClassiQ Web UI Tool classification report
8-11.5.2	Software tool qualification report	Tool qualification report	DTE	Tool qualification report
PS-SC	Confirmation measure reports	Tool confirmation review (recommended step at PS-SC)	DTE	Tool confirmation review
Proven in use argument				
8-14.5.2	Description of candidate for proven in use argument	Proven in use is not used by PS-SC. From expert view not practicable. If customer requirement, work is to be done according to standard. Project is responsible.	NA	NA
8-14.5.3	Proven in use analysis reports			

Interfacing an application that is out of scope of ISO 26262				
8-15.5.1	Base Vehicle Manufacturer or Supplier guideline.	This clause applies to T&B, where the base vehicles are delivered to body builders for further integration and hence generally not in PS-SC development scope	NA	NA
Integration of safety-related systems not developed according to ISO 26262				
8-16.5.1	Safety rationale	This clause applies to T&B, where a system not developed according to ISO 26262 is integrated. Not applicable to PS-SC, since the ECU/ VCU system are always developed according to ISO 26262 standard. Integration of system developed according to other standards are not in scope of PS-SC development	NA	NA

Overview of required Confirmation Reviews –

1. Hazard analysis and risk assessment – By Power Train Safety Team
2. Safety plan – By expert, review independency ASIL based
3. Item integration and test strategy – Product Verification Plan (PVP) review by Project Q champion/ Lead EPQ.
4. Validation plan – By OEM
5. Safety analyses and dependent failure analysis – Dept. Manager (FMEA), For FMEDA, FTA Refer the ProcLib, taken care by the method description.
6. Software tool criteria evaluation report and the software tool qualification report – By DTE (Domain Tool Expert)
7. The proven in use arguments – Not Used in PS-SC
8. Completeness of the safety case – Safety Case confirmation review by independent person (PSM) depending on ASIL based review independency
9. Functional safety audit and Assessment – By Safety auditor/assessor
10. Impact Analysis – Review by person (PSM) outside the department

11. Functional Safety Concept – By expert, review independency ASIL based
12. Technical Safety Concept – By expert, review independency ASIL based
13. Integration and test strategy – By expert, review independency ASIL based
14. Safety validation specification – By expert, review independency ASIL based

And in **Process -> Workflows and Activities -> Functional Safety (FuSa)Safety Activities in Project Preparation:** Functional Safety Assessment

5.2 Part 3: Concept phase

This part describes the ISO 26262 requirements during concept development of the item.
Customer requirements w.r.t. ISO26262 of the project are clarified during this phase.
If customer is aware of ISO 26262 and demands compliance, request following documents from the customer

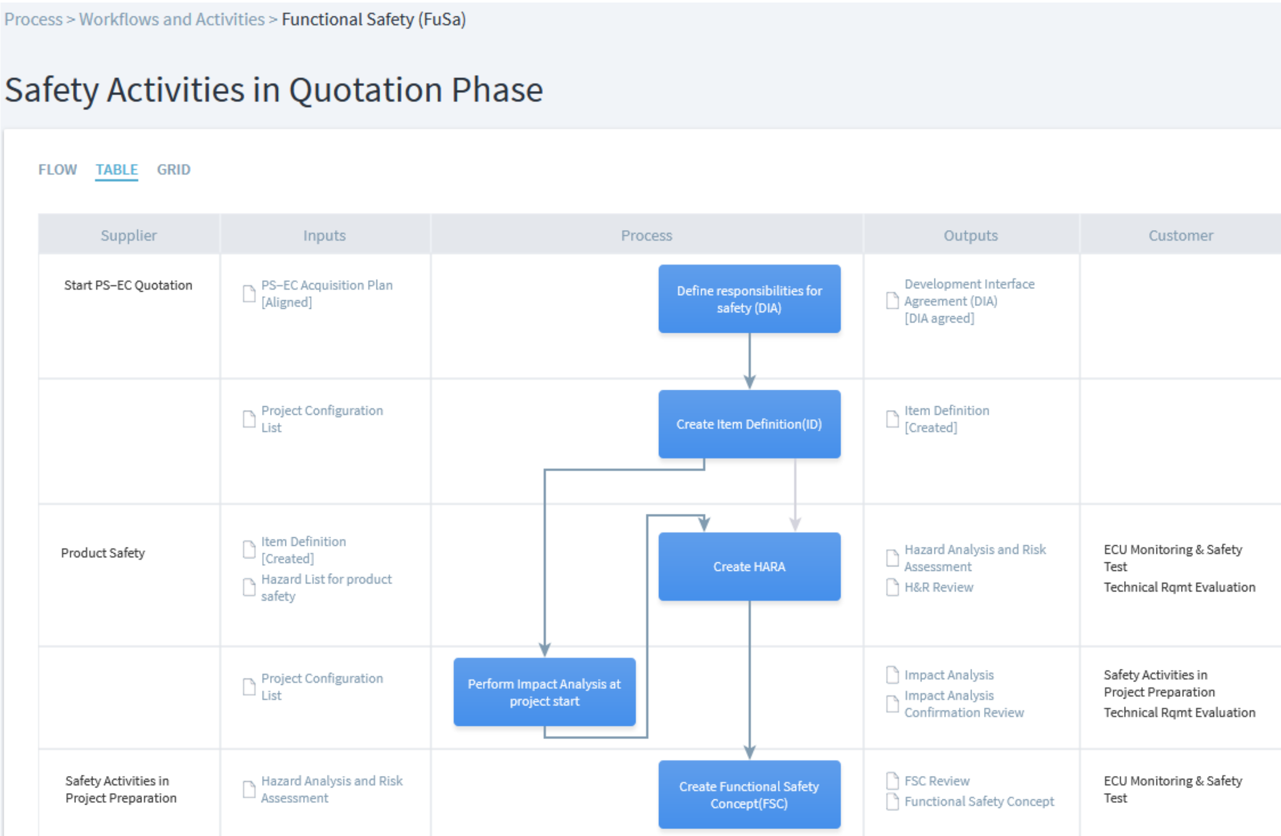
- Item Definition
- Hazard Analysis and Risk Assessment
- Safety Goals and ASIL Classification
- Safety Requirements
- Functional Safety Concept

If customer does not demand ISO 26262 compliance, implementation of ISO 26262 is done as per PS-SC strategy (Contact PSM for more details). In that case, platform documents available in the Process Library can be adapted for project.

Perform Impact Analysis to check if the project is ISO 26262 relevant or not. Explain PS-SC strategy to customer.

In PS-SC Process Library, requirements from Part 3 are specified in **Process -> Workflows and Activities -> Functional Safety (FuSa) -> Safety Activities in Quotation Phase**

[Safety Activities in Quotation Phase]



5.2.1 Mapping of all work products from Part 3 to Process Library

ISO section	Work product	Remarks	Responsible	Links/Examples*
Item Definition				
3-5.5	Item definition(ID)	In the responsibility of OEM. If OEM does not provide, use template from PLIB as basis.	OEM	[Item Definition]
Hazard analysis and risk assessment (H&R)				
3-6.5.1	Hazard analysis and risk assessment	In the responsibility of OEM. If OEM does not provide, use H&R from PLIB as basis, and document this in DIA	OEM	[H&R]

3-6.5.2	Verification review report of the H&R and the safety goals	Partly applicable as the H&R is delivered by the customer. - If H&R is not performed by the customer and RB internal H&R is tailored project specific, then a review with Sub-Cluster System Safety Platform (H&R-Review) is to be executed. Meeting minutes of the discussion with the review team may be stored in safety case as review evidence. - If H&R is provided by the customer and implementation is by RB, a verification review is required with the customer to avoid obvious incompleteness.	OEM	H&R Review
2-6.5.5	Confirmation measure reports (H&R)	Partly applicable as the H&R is delivered by the customer. - If H&R is not performed by the customer and RB internal H&R is tailored project specific, then a review with Sub-Cluster System Safety Platform (H&R-Review) is to be executed. Meeting minutes of the discussion with the review team may be stored in safety case as review evidence. Note: Verification review and Confirmation review of the RB internal H&R is already available. No additional reviews at project level necessary if used without changes. Also applies in case customer do not share the H&R details.	OEM	Review protocol is attachment to H&R in Plib

Functional safety concept				
3-7.5.1	Functional safety concept(FSC)	In the responsibility of OEM. If OEM does not provide, then 3 level monitoring concepts per RB platform recommendation can be used.	OEM	FSC
3-7.5.2	Verification report of the functional safety concept	Review protocol of generic FSC(If there is any change in the FSC, for which BOSCH is responsible, then only FSC Review is applicable)	OEM	FSC review

5.3 Part 4: Product development at the system level

Requirements from the system development are implemented in various processes in the PS-SC process library. Major requirements from Part 4 e.g. Technical Safety Requirements(TSR), Technical Safety Concept(TSC), TSR/TSC review, Item integration & testing plan and validation, Safety Analysis (FMEDA & FTA), Functional Safety Assessment (FSA) are specified in

Process -> Workflows and Activities -> Functional Safety (FuSa): Safety Activities in Project Design

And in **System Integration Test Processes:** Test Process of PS-SC Process Library.

5.3.1 Mapping of work products from Part 4 to Process Library

ISO section	Work product	Remarks	Responsible	Links/Examples
System integration and testing (Planning)				
Technical safety requirements				
4-6.5.1	Technical safety requirements (TSR) specification (component-level)	Safety Requirements at system level = RB safety requirements + customer specific safety requirements	SysArch	TSR
System Design				
4-6.5.2	Technical Safety Concept	TSC reference from platform are available in the Process Library.	SysArch	TSC

		However, Project specific adaptation necessary.		
4-6.5.3	System architectural design specification	Same as 4-6.5.2		LINK
4-6.5.4	Hardware software interface specification (HSI)	ISO 26262 requirements are mapped in the position paper and it is the bare minimum requirement from the standard. The work products can be enhanced to suit customer specific needs at the discretion of the PSM. (e.g. an HSI in Doors)	PSM	HSI
4-6.5.5	Specification of requirements for production, operation, service and decommissioning	Projects to provide only special characteristic, part of Technical Customer Documentation (TCD/ TKU)	PjM ECU	LINK
4-6.5.6	Verification report for system architectural design, the hardware-software interface (HSI) specification, the specification of requirements for production, operation, service and decommissioning, and the technical safety concept	System verification report (refined)	PjM ECU	LINK
4-6.5.7	Safety analysis reports	Reports from Safety Analysis done in project	PSM	Safety Analysis

		System safety FMEA, FMEDA and FTA		
2-6.5.5	Confirmation Review safety analysis	Confirmation Review of System safety FMEA by dept. manager, PS-SC/QMM, Refer ProcLib template- Confirmation review of FMEDA and FTA	PSM	Safety Analysis Conformation Review (FMEA,FMEDA, FTA)
Item integration and testing				
4-7.5.1	integration and test strategy	PVP covers most of the tests at the PVER level and accepted as the 'Integration and Test strategy. Review by the Lead EPQ at QGC0 is accepted as evidence of confirmation review.	PjM ECU, Lead EPQ	LINK PVP
4-7.5.2	Integration and Test reports	**For ECU ASIL-B/QM:** - ECU-MST: a) Level 1: only part which is not covered by L2 b) Level 2/3: complete - SRD - PVER-Conf: I/O testing - Task in PVER Version Creation: "Measure SW resources" **For ECU ASIL-C/QM or ASIL-D/QM:** same system test as for ASIL-B/QM and additionally: Requirements from stress tests and resource usage tests.	PjM ECU	TestSpecification ECU-MST System Qualification Test (SysQT) SRD PVER-Conf Measure SW resources

		- COM-Test (part of Robustness-Tests) - I/O-Test (part of Robustness-Tests) - BFT / Runtime Test (BFT_3_3) - VIVA (to be ordered by OEM) -FT, alternatively OT: process Step “Test dynamic SW resources” **Note:** Definition of stress conditions project specific under OEM responsibility -> e.g. testing done by Bosch, assigned by OEM **Note:** I/O-Test can be tailored (change based testing) for ASIL C but **not** for ASIL D		Robustness Test (COM-Test) Robustness Test (I/O-Test) BFT VIVA FT / OT test report
Safety validation				
4-8.5.1	Safety validation specification including safety validation environment description	The overall responsibility of the validation on the OEM side. The OEM can use several ECU-MST-Tests cases for Validation.	OEM	NA
4-8.5.2	Safety Validation report	The overall responsibility of the validation on the OEM side.	OEM	NA

5.4 Part 5 & 8: Product development at the hardware level & Supporting Processes

A link to a guided tour for ISO 26262 implementation for HW is available in process library, see Functional Safety view.

5.4.1 Mapping of work products from Part 5 & 8 to Process Library

ISO section	Work product	Remarks	Responsible	Links / Examples
Specification of hardware safety requirements				
5-6.5.1	Hardware safety requirements specification (including test and qualification criteria)	Basic HW safety requirements (Safety RB Requirements HW) + Requirements from req. management tool (HW safety requirements specification	PjM HW	HW safety req spec HW Safety Requirements to use DOORs (mandatory) Method description: HW_RA_Safety_ Method
5-6.5.2	Hardware-software interface specification (refined)	See 4-6.5.3	PSM	HSI
5-6.5.3	Hardware safety requirements verification report	Review of the Hardware safety requirements to check consistency with the TSC, SDS and SWS	PjM HW	Perform HW Safety Req Review xCU Review Protocol HW: Review of Req. are documented in "HW Safety

				requirements Review" (ID of HW-Review tool, report can be exported)
Hardware design				
5-7.5.1	Hardware design specification	HW Architectural Design • HW Requirement Specification: HW Safety requirements are allocated to HW Modules • Modul list HW: HW Modules are specified including their Safety requirements HW Detail Design: xCU schematic, HW Module Specification and Datasheets	HW DEV ENG	Module List HW xCU Circut diagram Circuit Diagram - Integrating all modules from module list Original file is saved in eADM, pdf version to be saved in safety case.
5-7.5.2	Hardware safety analysis report	Safety Analysis: FMEDA, FTA, FMEA & DFA • Product FMEA with inputs from HW Module FMEA element.	ECU PjM, SWSD, HW DEV MOD, PSM	Safety Analysis HW Module FMEA Element Design FMEA ECU
5-7.5.3	Hardware design verification report	Review protocol Integration Review HW (ID in Review DB)	PjM HW	Perform the Final Review (Layout)

5-7.5.4	Specification of requirements related to production, operation, service and decommissioning	Projects to provide only special characteristics (If applicable)	PjM ECU + PjM HW	TCD/TKU
Evaluation of the hardware architectural metrics (mandatory for ASIL C/D)				
5-8.5.1	Analysis of the effectiveness of the architecture of the item to cope with the random hardware failures (HW Architectural Metric)	Quantitative Analysis: FMEDA for SPFM and LFM.	PSM	FMEDA
5-8.5.2	Verification review report of evaluation of the effectiveness of the architecture of the item to cope with the random hardware failures	FMEDA (SPFM, LFM) is verified by review. For review template refer to the Link	PSM	Safety Analyses Confirmation Review
Evaluation of violation of the safety goal due to random HW failures (mandatory for ASIL C/D)				
5-9.5.1	Analysis of safety goal violations due to random hardware failures (HW Reliability Metric	Quantitative Analysis: FMEDA for PMHF.	PSM	FMEDA
5-9.5.2	Specification of dedicated measures for hardware, if needed, including the rationale regarding the effectiveness of the dedicated measures	See FMEDA process steps	PSM	FMEDA
5-9.5.3	Verification review report of evaluation of safety goal violations due to	FMEDA (PMHF) is verified by review. For review template refer to the Link	PSM	Safety Analyses Confirmation Review

	random hardware failures			
HW Integration and testing				
5-10.5.1	Hardware integration and verification specification	Test case specification is part of HW_TST_xx_TPS _Method. Refer to link (all Test case specifications are listed, e.g. HW_TST_VERDICT_TPS_Method). Additionally, for ASIL C and D - Fault injection test (refer to MD HW TST FAQ)	PjM HW	Test Specification HW xCU HW TST Fault Injection HW strategy
5-10.5.2	Hardware integration and testing report	Test report of Tests from Test specification mentioned in 5-10.5.1		Test Report HW xCU
Qualification of hardware components				
8-13.5.1	Hardware elements evaluation plan		NA	
8-13.5.2	Hardware element test plan	Relevant only for intermediate components (HW Modules) which are not	NA	

		a part of the part No. 5 of ISO26262.		
8-13.5.3	Hardware element evaluation report for hardware elements	Relevant only for intermediate components (HW Modules) which are not a part of the part No. 5 of ISO26262.	NA	

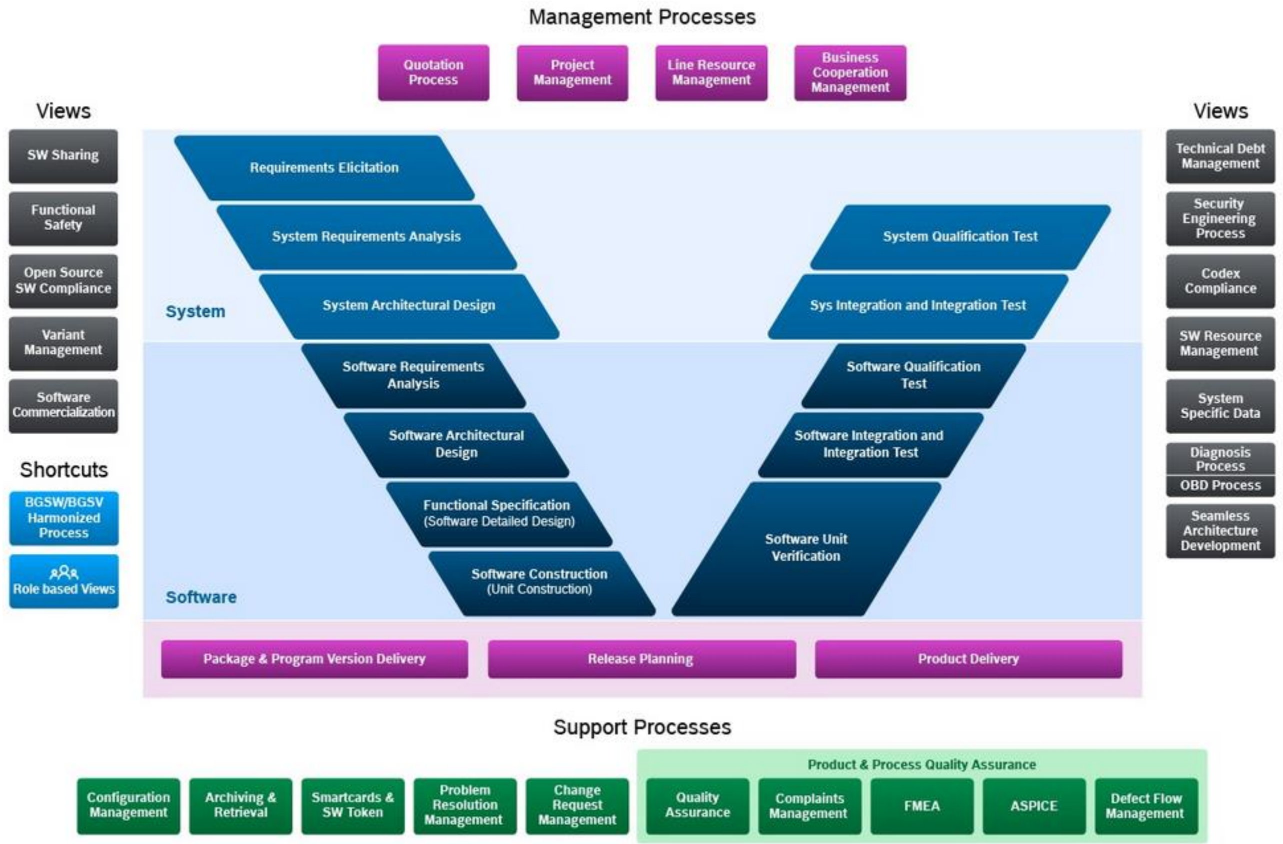
5.5 Part 6 & 8: Product development at the software level & Supporting Processes

The requirements from part 6 can be met by following the software development processes of the PS-SC processes library.

Process Group: SW Product Development

The software development sub-phases and supporting processes are initiated by determining the appropriate methods to comply with the requirements and their respective ASIL. The analysis of these methods is given in the section 7.3.3 – Position paper - Software Development.

Requirement based development (PTSA 2.x)



5.5.1 Mapping of work products from Part 6 & 8 to Process Library

ISO Section	Work product	Remarks	Responsible	Links/Examples
Planning				
6-5.5.1	Documentation of the software development environment	Design and coding guidelines for modelling and programming languages	PjM ECU	General and monitoring coding guideline DGS ASCET modelling guideline ASCET Resources
Software safety requirements (FN-D Mo and SWSD with support from PSM and FS-Res Mo)				

6-6.5.1	Software safety requirements specification	Basic SW safety requirements (RB SW Safety Requirements) + Requirements transmitted from requirement management tool. Project specific safety requirements identified using Architecture list(hotlist)		SW Safety Requirement Customer Requirements
6-6.5.2	HW-SW interface specification (HSI) refined	HSI shall describe each safety-related dependency between hardware and software		HSI
6-6.5.3	Software verification report	The review of HSI specification is handled in HSI position paper.		HSI
SW architectural Design (FN-D Mo with support from FS Res Mo)				
6-7.5.1	Software architectural design specification	How an architectural design specification requirement of ISO26262 are satisfied is documented and is available in a position paper.		mdSW_PP_FS_SWARCH.pdf
6-7.5.2	Safety analysis report	FMEA for Monitoring relevant functions –> FMEA-Mo		Process : FMEA
6-7.5.3	Dependent failures analysis report	The FMEA for MoCPFC_Co, MoCMem_Co, GC_MO : MoFVehIfc (External data exchange) and freedom of interference pdf		Process : FMEA Freedom from Interference (MDG1)
6-7.5.4	Software verification report (refined)	Position paper in place for SW Verification		Position paper For ASIL C& D: Control flow analysis and/or Data flow analysis

				necessary, apart from ASIL B work products
Software unit design and implementation (FN-D Mo, SW-D Mo)				
6-8.5.1	Software unit design specification	Based on the software architectural design, the detailed design of the software units is developed.		Function Specification Process : Function Spec and Documentation Design guidelines ASIL B and above -> aaSW_Common_DGS_Monitoring_SW_Guidelines_en.doc
6-8.5.2	Software unit implementation	Based on the detailed design, the implementation of the software units is made.		Process : Code Object, ASCET File Coding guidelines ASIL B and above -> Common_DGS_Monitoring_SW_Guidelines_Checklist.xls Common_DGS_Monitoring_FIT_Guidelines.doc (note – accessible only by Mo-Teams)
SW unit test (FN-D Mo, SW-D Mo)				
6-9.5.1	Software verification specification	(SCC, Code Review and UT) Test specification		Process : SCC Code Review URT
6-9.5.2	Software verification report (refined)	(SCC, Code Review and UT) Test reports		Process : UV-Finished
Software integration and verification (FN-D Mo, SWSD, SWPD, FA-Mo)				
6-10.5.1	Software verification	PVER Integration Test specification,		Process :

	specification (refined)	ECU MST Specification, Resource measurement at package level.		Requirements-based test => - SwQT Interface test => - SwQT Fault injection test => - SwQT with reference to test design methods Resource usage test => Program Version Creation (task Resource measurement) Resource Estimation and Tracking Back-to-back comparison test between model and code, if applicable => Not followed for monitoring software. Verification of the control flow and data flow => - UT - Static tests on package level (SCC-BC) - or Static check at PVER level (PVER-I) Static Code Analysis => - Static tests on package level (SCC-BC) - or Static check at PVER level (PVER-I) Static analyses based on abstract interpretation => - AOOB
6-10.5.2	Embedded Software	Program Version (PVER) build output.		PVER

6-10.5.3	Software verification report (verification of integrated software)	Package Integration Test report, Robustness test report , ECU MST report		Test reports from the steps mentioned in 6-10.5.1
Verification of software safety requirements (FN-D Mo, FA-Mo)				
6-11.5.1	Software verification specification (refined)	SW QT a) project specific entry evaluation (Impact Analysis) b) SWQT specification		Test Specification SwQT
6-11.5.2	Software verification report (refined)	Test report SWQT		Test Report SW QT **For ASIL C & D:** Function coverage and/or Call coverage report necessary, apart from ASIL B work products.
Annex C Software configuration(FN-D Mo, SWSD, SWPD**)**				
6-C.5.1	Configuration data specification	Specification for - System Constants - Compiler Settings => is described in Data Version DST DST = Data Version = Program Version (PVER) [Code] + specific (project, product) set of parameters = PVER + a2l		Process : Dataset Version (DST) Program Version => PVER
6-C.5.2	Calibration data specification	Specification for - Parameter range - Quantization		Process : PVER-Calibration data specification

		=> is described in A2L File		use of the safety-relevant calibration data
6-C.5.3	Configuration data resulting from safety requirement	Determination of - System Constants - Compiler Settings => are done/configured in PVER		Process : Program Version (PVER)
6-C.5.4	Calibration data resulting from safety requirement	calibrated Hex-File as a part of Data Version DST Series data set (HEX file) containing - Code words (CW) - Calibration data (label)		Process: Series dataset (Hex file)
6-C.5.5	Calibration data Verification specification	Create Test Cases for ECU-MST		Process : ECU MST
6-C.5.6	Calibration data Verification report	ECU-MST: a) Results from eMFace or equivalent Dependant from the result of the project specific entry analysis (see 6-11.5.2): b) Protocol data check L1 (safety-related input 4 L2 data) c) Protocol data check L2/3 d) Test report L1 (of QM safety requirements) e) Test report L2/3		Process : ECU MST
6-C.5.7	Software Architectural design specification (refined)	see 6-7.5.1		
6-C.5.8	Documentation of the software	see 6-5.5.1		

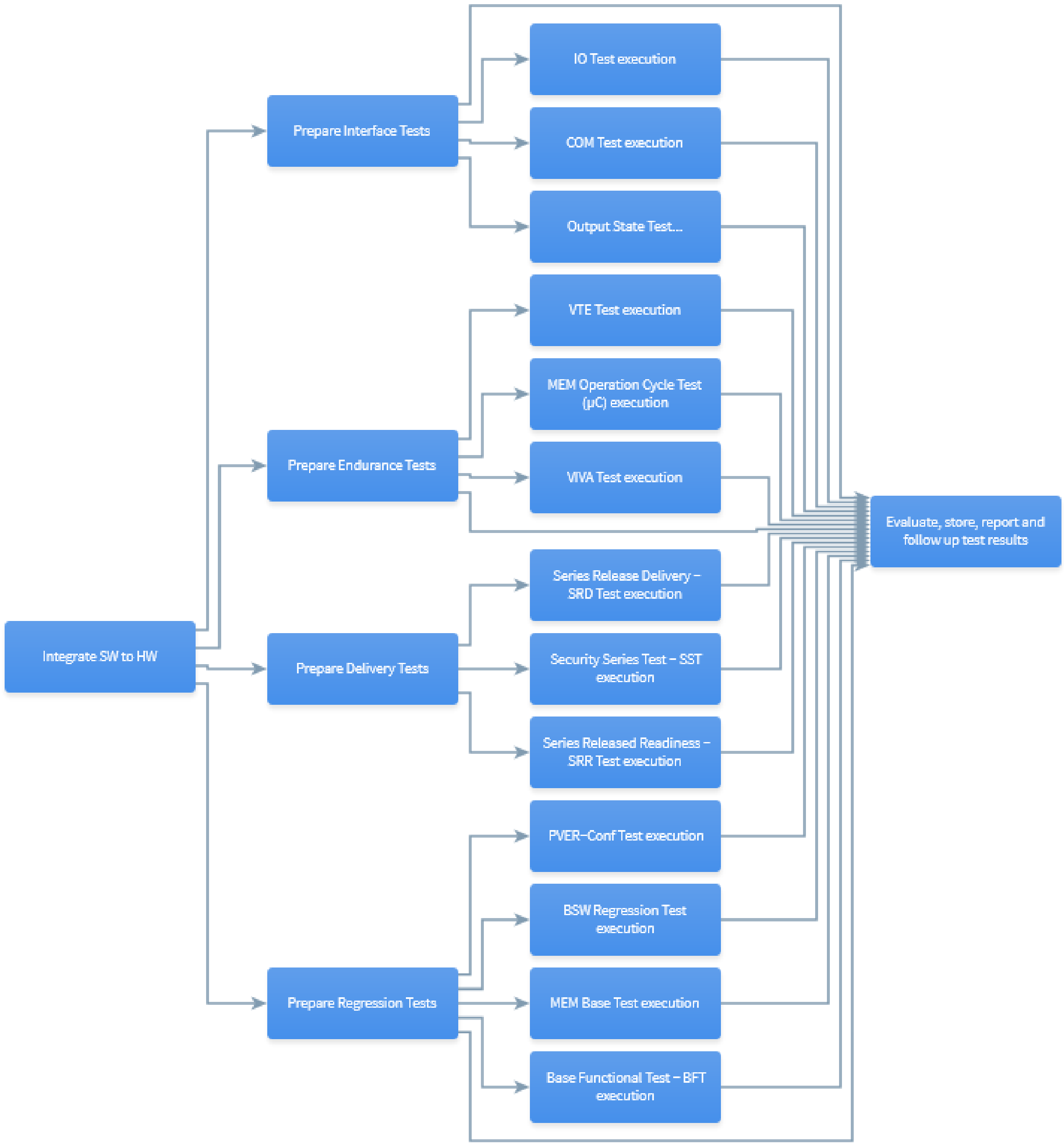
	development environment (refined)			

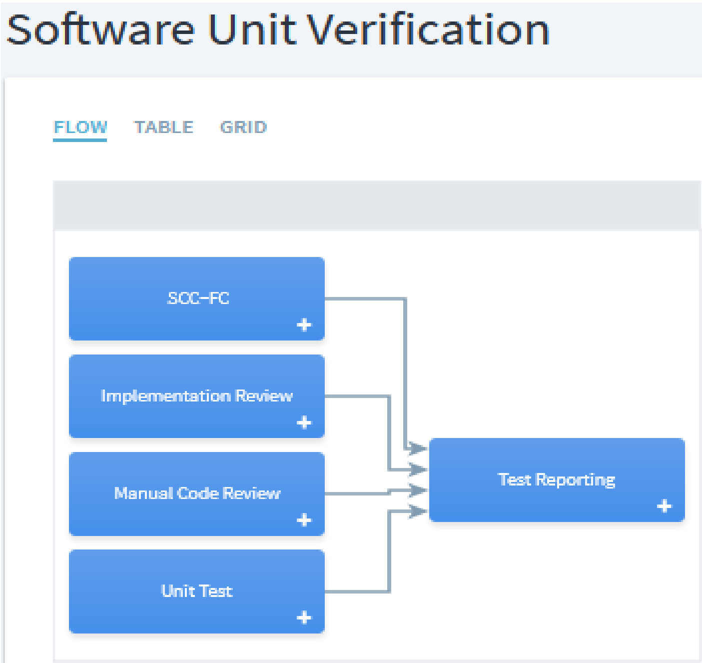
* Examples of the work products are available in a central folder (LINK) accessible only by PSMs. Kindly contact them in case you want to refer them. The links are mainly for the PTSA 1.x based SW product development. See SYS and SW position paper for PTSA 2.x work products mapping.

6 Requirements on Tests

CE Test Processes & SW Test Processes

In General, PS-SC CE test processes are sufficient to fulfill the ASIL test requirements of ISO 26262.


IO Test execution
Prepare Interface Tests
COM Test execution
Output State Test...
VTE Test execution
MEM Operation Cycle Test (µC) execution
Prepare Endurance Tests
VIVA Test execution
Evaluate, store, report and follow up test results
Series Release Delivery – SRD Test execution
Integrate SW to HW
Prepare Delivery Tests
Security Series Test – SST execution
Series Released Readiness – SRR Test execution
PVER-Conf Test execution
BSW Regression Test execution
Prepare Regression Tests
MEM Base Test execution
Base Functional Test – BFT execution



HW Test Processes

PS-SC HW test processes are sufficient to fulfill the ASIL C & D test requirements of ISO26262. Fault Injection test has to be additionally done for ASIL C. Refer position paper link for HW Fault injection testing.

7 Difference between ASIL B and ASIL C/D

The major differences between ASIL B and ASIL C/D are

- Additional processes
- Work products
- Methods followed during the product development
- Test Methods

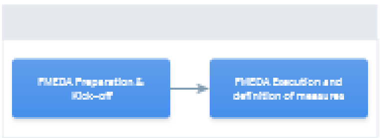
7.1 Additional processes for ASIL C/D

Additional process steps that are mandatory for ASIL C & ASIL D development

Quantitative Analysis Process (FMEDA) → FMEDA

FMEDA

FLOW TABLE GRID



DESCRIPTION

IN GENERAL

The Purpose of the Workflow ***FMEDA*** :

A quantitative analysis is requested by ISO26262 for an ASIL classification higher or equal ASIL C or on customer specific request.
It has to be demonstrated that the specific safety goals can be fulfilled and are within the requested limits.

Variety of methods exists for quantitative analysis.

The Failure Mode Effects and Diagnostic Analysis (FMEDA) is the recommended analytical method of preventive quality management in PS-EC product development , and is carried out according to

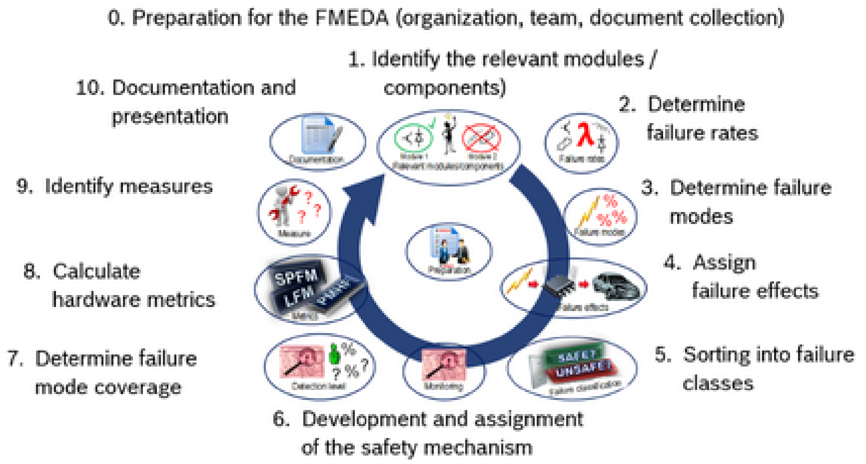
Hardware development and system development are the areas of application of the FMEDA.

It helps to calculate metrics for safety-relevant electronic systems as required in ISO26262.

It considers individual errors of components and elements in the fault detection in the system.

The FMEDA has to be carried out during the development phase.

10 steps for compiling an FMEDA



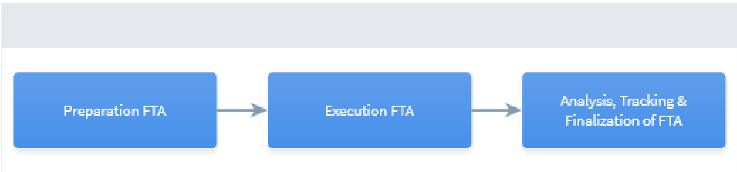
WORKFLOW ACCEPTANCE CRITERIA

FMEDA is mandatory for projects with safety requirements rated ASIL C, ASIL D, ASIL X(C), ASIL X(D)
FMEDA can also be useful for safety requirements rated ASIL B, for design of new systems or architecture or similar use cases.

- Qualitative Analysis Process (FTA) → [FTA](FTA)

FTA

FLOW TABLE GRID



DESCRIPTION

IN GENERAL

The Purpose of the Workflow ***FTA*** is to conduct a deductive analysis

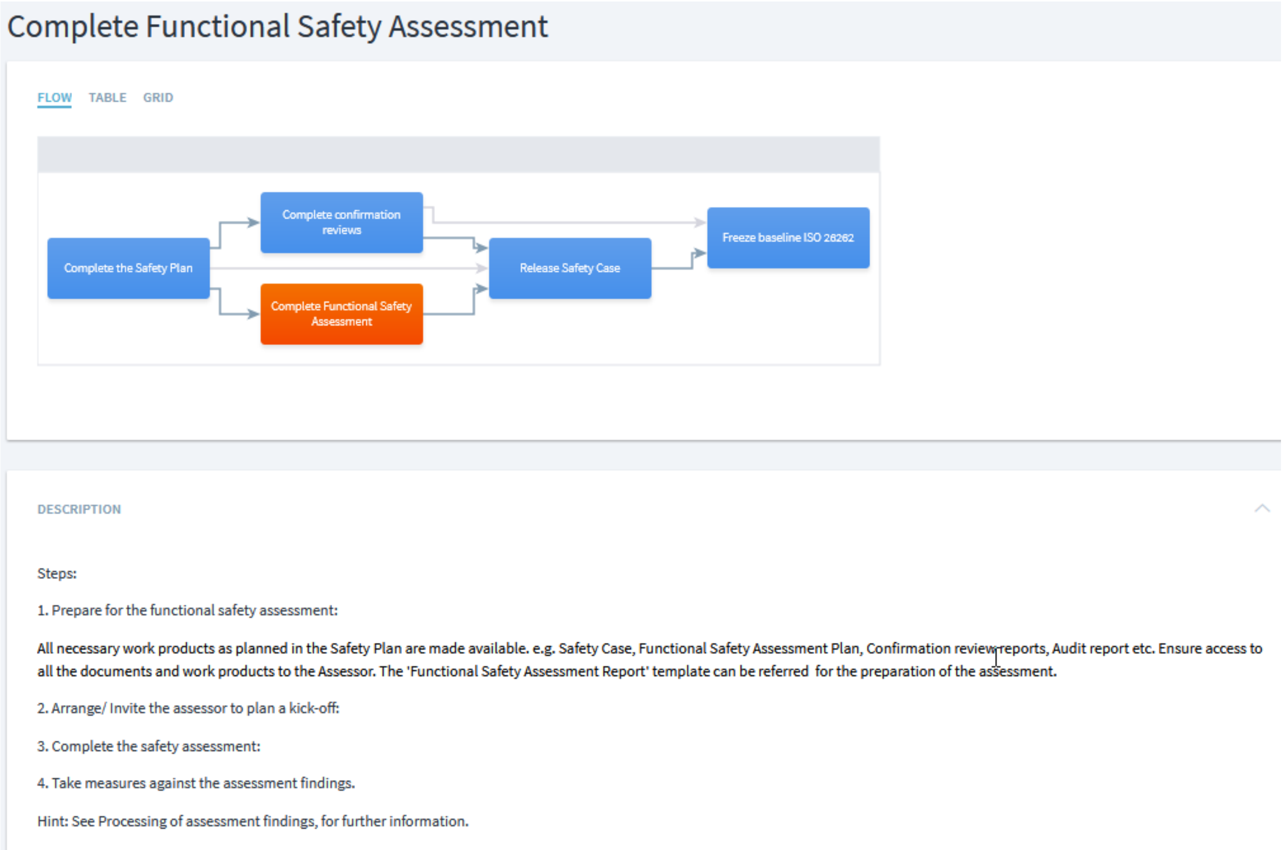
A deductive analysis on the system architectural design is requested by ISO 26262 for ASIL C and D projects, or on customer request.
At PS-EC, Fault Tree Analysis is performed as a qualitative analysis method, to show that the system design is sufficient to fulfil the specific safety goals.
FTA should be completed before the HW QGC2/QGP1, so as to accomadate any design changes if necessary.

WORKFLOW ACCEPTANCE CRITERIA

The Workflow ***FTA*** is completed if all below listed workflow acceptance criteria are done.

- FTA report available
- Confirmation review completed

- Functional Safety Assessment Process → [FSA]



7.2 Work Products

Work products which are output from process mentioned in section 7.1 e.g. FMEDA file, Functional Safety Assessment Report, FTA report etc.

From SW development - Control flow analysis and/or Data flow analysis reports, Function coverage and Call coverage report.

From HW development - HW Fault Injection Test Report

7.3 Methods followed during the product development

There are different methods e.g. test and review methods, recommended by ISO26262 standard during the product development for ASIL C and above. Analysis of these methods, conformity and justification are given below. These are indicated by the corresponding position papers.

7.3.1 Position paper - System Development

[Position Paper System]

7.3.2 Position paper - Hardware Development

HW FuSa PP Hardware Method

7.3.3 Position paper - Software Development

Position papers for SW development

7.4 Test Methods

See **6. Requirements on Tests**

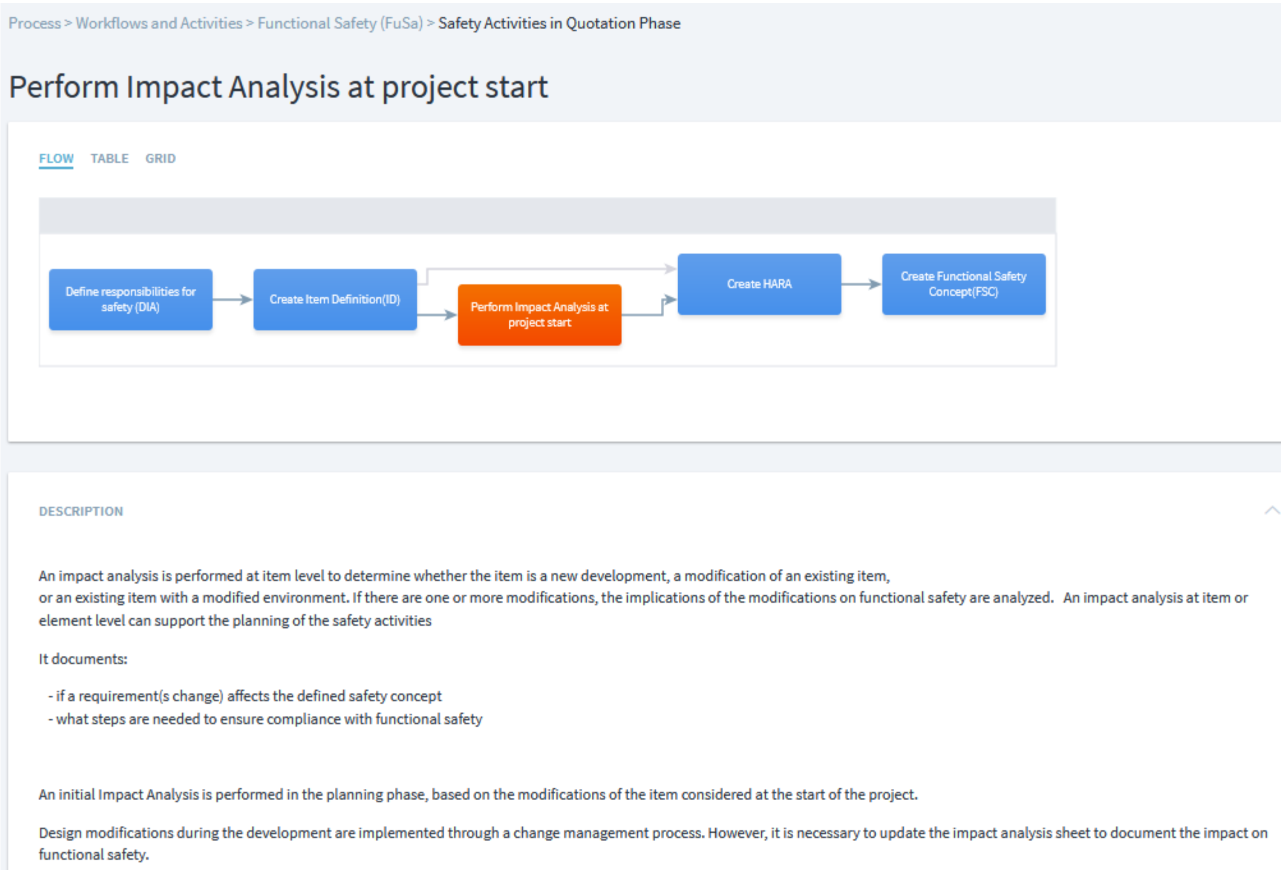
8 Product & Process Quality Assurance and Product Release

Process Group: Product & Process Quality Assurance –

Quality Gate assessment question for Functional Safety ISO26262 is implemented in QGx questionnaire. The rating of this question corresponds to the total outcome of Impact Analysis Checklist. It is used to check if the required activities according to ISO 26262 depending on different roles are completed or not. The checklist is available at

Process Group: Quotation Process > Kick-off

Process: Impact Analysis



Process Step: Prepare Quality Gate meeting

Product Release

Further, compliance to ISO26262 is also checked during the Technical Product Release.

Product Delivery > Technical Product Release > Check for completeness of safety related tasks

Process > Workflows and Activities > Product Delivery

Technical Product Release Preparation

FLOW TABLE GRID

DESCRIPTION

IN GENERAL

In this process it is determined, whether the product can be released for series production from a technical and quality point of view. This is done by evaluating the project risk list and the PVP.

Tailoring:

- Tailoring is possible depending on the intended use of the PVER - see Product release Tailoring reference below in Guidance , regarding QGC4 as required input document and regarding approval of the PVP

Good Practice:

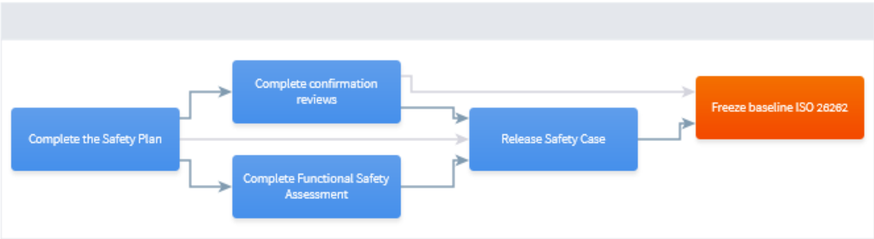
- To evaluate the technical product release from a system point of view (also for D-Samples) the following list can be used: Checklist for SOP Release reference in below Guidance

WORKFLOW ACCEPTANCE CRITERIA

- Get confirmation for correct handling of vehicle variants
- Update TCD (TKU)
- Check for completeness of safety related tasks
- Evaluate risk list of the project
- Evaluate and update the PVP
- Perform QGC4 and QGF meeting

Freeze baseline ISO 26262

FLOW TABLE GRID



DESCRIPTION

IN GENERAL

This Purpose of this activity is to perform the baseline of the functional safety work products and to provide the ISO 26262 baseline.

The project safety manager collects the valid versions of the required functional safety work products the ISO 26262 Baseline (Baseline Template).
- apply the applicable "Configuration Management" process, refer to method description "ECU Configuration Management".

The Purpose of this activity is to verify and freeze the functional safety baseline.

Verification includes checks that the configured items are consistent with each other and the baseline plan, e.g., that the correct work products are included, that all planned changes/ requirements for the series delivery are included.

Freeze the baseline according to the used baseline tool (e.g. workflow in SharePoint) and release the functional safety baseline for product release.
- apply the applicable "Configuration Management" Process

Baselining for Projects using SharePoint:
For projects using SharePoint as document management system, baselining is done in SharePoint according to the specifications in the "Project Specific CM Plan".

ACTIVITY ACCEPTANCE CRITERIA

The activity freeze baseline ISO26262 is completed if all below listed activity acceptance criteria are done.

- ISO 26262 baseline is reviewed and released

9 Handling of Safety Requirements

Safety Requirements are available at system, software and hardware level, they are called as Functional Safety Requirements, Technical Safety Requirements, Software Safety Requirements and Hardware Safety Requirements respectively.

For each of these requirements, RB platform safety requirements are defined and are available in Process Library.

SYS SW - ECU (RbD 2.x) > System Requirement Analysis

Workproduct > Requirements > Requirements Impact Analysis Result

Process > Work Products > Requirements

Requirements Impact Analysis Result

PROCESSES DEPENDENCIES STATES

DESCRIPTION

IN GENERAL

The Work Product ***Requirements Impact Analysis Result*** describes the engineering impact[1] a change request will have on the product under development.

It describes the impact on the following - minimum but not limited to - engineering domains and topics:

- Engineering Domains
  - Requirements Elicitation
    - Stakeholder Requirements Baseline
  - System Development
    - System Context
    - System Requirements
    - System Architecture Design
    - System Test
  - Software Development
    - Software Requirements
    - Software Architecture Design
    - Function Specification (Software Detailed Design)
    - Software Construction (Unit Construction)
    - Software Test
  - Hardware Development
    - Hardware Requirements
    - Hardware Architecture Design
  - Calibration Development
    - Calibration Data
- Engineering Topics
  - Compliance
  - Functional and Product Safety
  - Exhaust Emission
  - On-Board-Diagnosis (OBD)
  - Security
  - Aftermarket Functions
  - Software Licensing

PAY PARTICULAR ATTENTION TO THE FOLLOWING STATES

- [documented by FD Team]

  State [documented by FD Team] is assigned to the *Requirements Impact Analysis Result* if all of the following criteria are fulfilled:
  - The impact of the change request on the corresponding functional development is documented completely and consistently

- [filed by Issue SW]

  State [filed in Issue SW] is assigned to the *Requirements Impact Analysis Result* if all of the following criteria are fulfilled:
  - The impact of the change request on all affected engineering domains of the product is documented completely and consistently, including the results of the analysis from the FD Team

RbD2 - Engineering Impact Checklist

FILES

NAME	AUTHOR	LAST CHANGE	STATE
Engineering_Impact_Chklst	Maumita Datta (MS/EMP-PS-PJ-MB)	07/15/2022 11:47 AM	

DESCRIPTION

IN GENERAL

The Guidance ***RbD2 - Engineering Impact Checklist*** describes a set of basic questions as basis for an impact analysis in system and/or software development (in RbD 2.x). The focus of the questions is on evaluation of the impact ("What will change?") and the associated risks ("What could go wrong?"). In short, the Guidance ***RbD2 - Engineering Impact Checklist*** shall support to identify the impact and risks.

Technical Safety Requirements (TSR) –

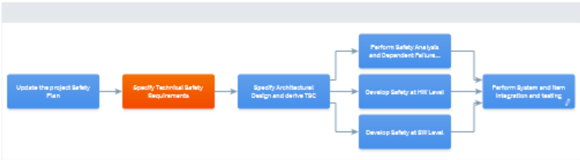
These are the safety requirements at system(ECU) level. Projects shall take this as a base and adapt it project specifically. If there are any project specific safety requirements, project shall add them to TSR and shall review them with required participants.

Management Processes > Project Management > Project Design > Carry out Functional Safety (ISO26262) specific activities

Process > Workflows and Activities > Functional Safety (FuSa) > **Safety Activities in Project Design**

Specify Technical Safety Requirements

FLOW TABLE GRID



DESCRIPTION

IN GENERAL

The purpose of this Activity Technical Safety Requirements is to specify the technical implementation of the functional safety requirements at their respective hierarchical level; considering both the item definition and the system architectural design, and addressing the detection of latent failures, fault avoidance, safety integrity and operation and service aspects.

These are the system (control unit) level safety requirements derived from functional safety concept and are relevant for functional safety ISO 26262.

These could be RB Safety Requirements or New Safety Requirements. If there are any new safety requirements e.g. customer specific, then add it to project specific Technical Safety Requirements

Technical Safety Requirements for MDG1 ECU platform can be found in DOORS

Handling of ASIL attributes of requirements are described in 'ASIL attribute for requirement' document.

ACTIVITY ACCEPTANCE CRITERIA

The Activity is completed if Technical safety requirements are correctly specified, mapped to the Functional Safety Requirements and HW/SW requirements

ADDITIONAL DOWNLOADS

- ASIL Decomposition Method
  Arun Kumar Muthu (MS/PJ-ProcM-PS) - 06/28/2022 11:13 AM
- Criteria for Coexistence
  Arun Kumar Muthu (MS/PJ-ProcM-PS) - 06/28/2022 11:20 AM
- DOORS
  Srikanth Veeraiah (MS/EMP2-PS) - 04/13/2022 9:01 AM
- FuSa_TSC_property_of_technical_independence_en_Method.pdf
  Arun Kumar Muthu (MS/PJ-ProcM-PS) - 05/18/2022 12:40 PM
- How to- ASIL attribute for requirements
  Arun Kumar Muthu (MS/PJ-ProcM-PS) - 05/11/2022 2:07 PM
- MDG1 platform technical safety requirements (in DOORS)
  Choorakunnel Toms Rajesh (MS/PJ-FS2-PS) - 03/10/2022 11:01 AM

Technical safety requirements:

Technical Safety Requirements

PROCESSES DEPENDENCIES STATES

DESCRIPTION

IN GENERAL

The Work Product ***Technical Safety Requirement*** describes ...

The technical safety requirements specify the technical implementation of the functional safety requirements at their respective hierarchical level; considering both the item definition and the system architectural design, and addressing the detection of latent failures, fault avoidance, safety integrity and operation and service aspects.

These are the system (control unit) level safety requirements derived from functional safety concept and are relevant for functional safety ISO26262.

These could be RB Safety Requirements or New Safety Requirements.

Note – Requirement Management for TSR is done according to project specific CM plan.

HW Safety Requirements –

These are the Hardware level safety requirements derived from Technical Safety Requirements. Projects shall take Safety RB Requirement HW as a base and adapt it project specific.

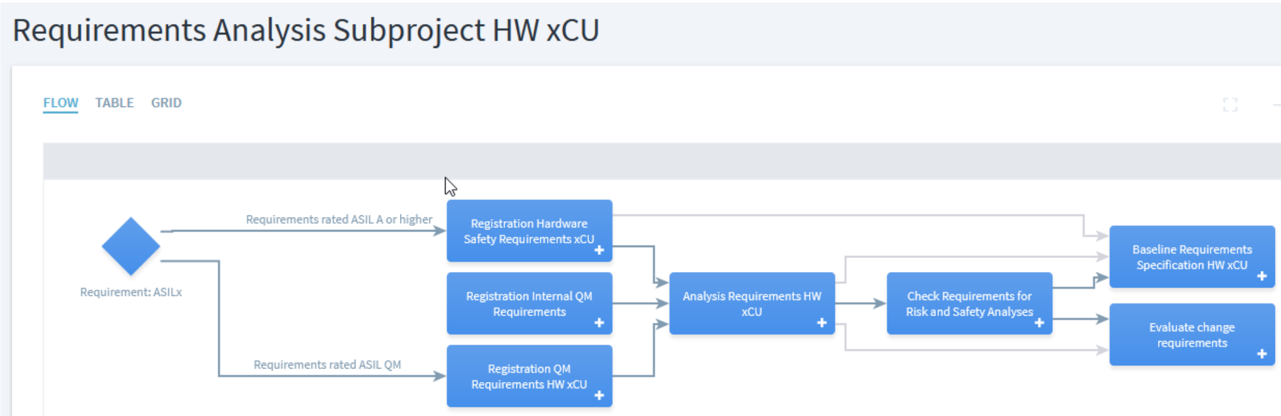
New **Issue HW** shall be created to handle HW Safety Requirement.

Software Safety Requirements –

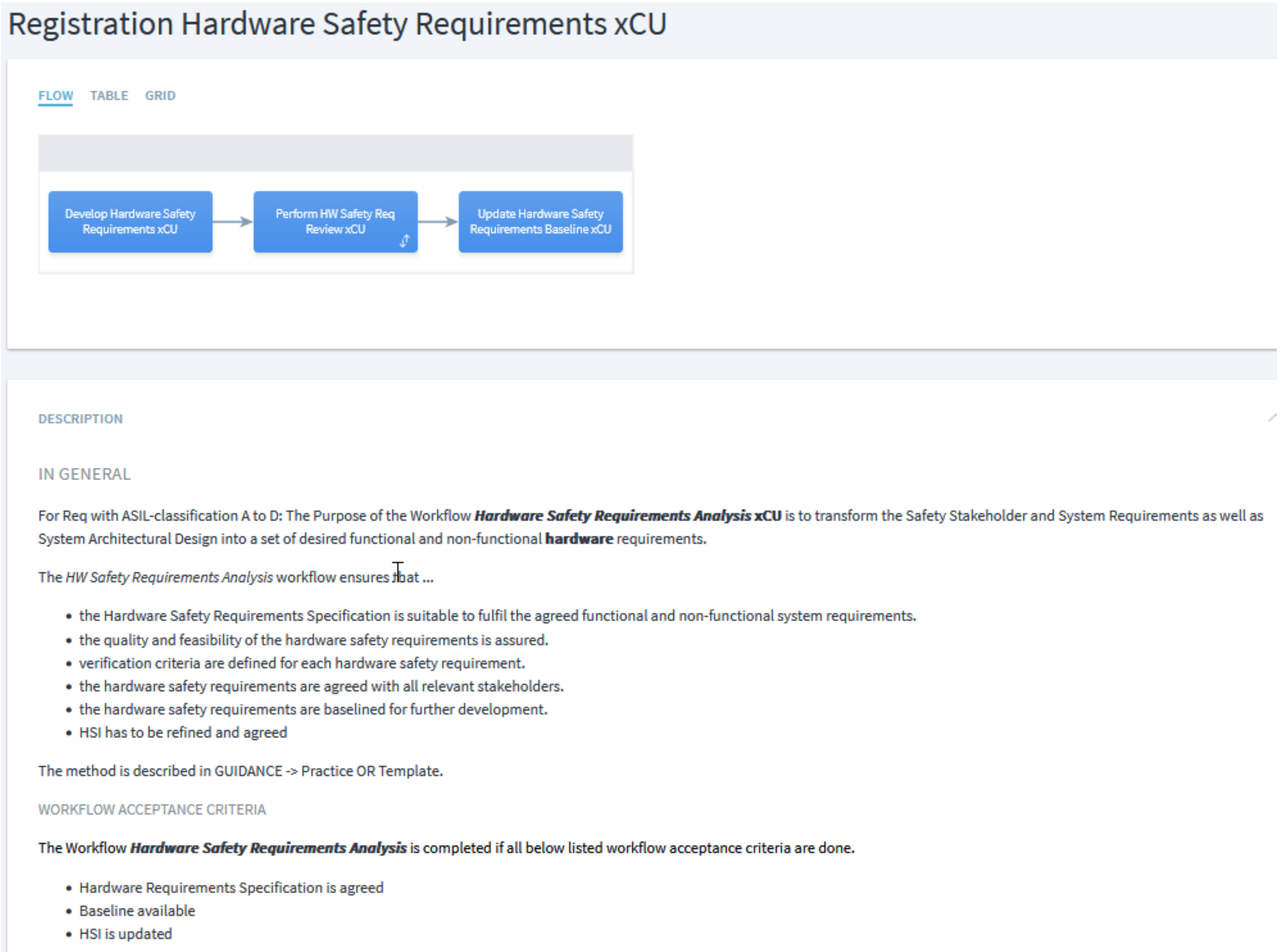
These are the software level safety requirements derived from Technical Safety Requirements. Projects shall take RB SW Safety Requirement as a base and adapt it project specific.

New **Issue SW** shall be created to handle SW Safety Requirement and analysed.

HW Product Development > Requirements Analysis Subproject HW xCU> **[Registration Hardware Safety Requirements xCU]**



Artifact: HW safety requirements specification



10 Work Flow for safety requirements

Handled as per the recommended tools in PLIB

Example: DOORS, RQ1

11 Suppliers Management

The customer and the suppliers have to jointly comply with the requirements specified in ISO 26262 part 8-5 **Interfaces within distributed developments**.
These requirements are specified for PS-SC External Suppliers in Process LIB in **Process Group: Management Process >** Business Cooperation Management

Subcontract Management

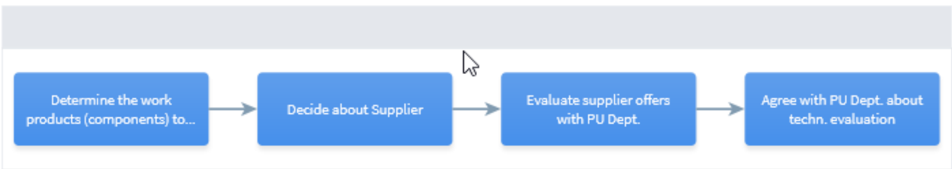
FLOW TABLE GRID

Supplier	Inputs	Process	Outputs	Customer
	External commissioning of work products necessary	Subcontract Management Software +	Acceptance Declaration Project Level Agreement Supplier order	
Software Requirements Analysis	Software Requirement [agreed]	Supplier Monitoring +	Package Version [released]	

[Process > Workflows and Activities > Subcontract Management]

Subcontract Management Software

FLOW TABLE GRID



DESCRIPTION

IN GENERAL

The Purpose of the Workflow ***Subcontract Management Software*** is to determine the process between PS-EC with external and RB internal suppliers acting as subcontractors for software development.

WORKFLOW ACCEPTANCE CRITERIA

The Workflow ***Subcontract Management Software*** is completed if all below listed workflow acceptance criteria are done.

- BCM Project Level Agreement completed and signed
- BCM External Commissioning prepared ready to sign
- Project Level Agreement completed and signed

Within this Workflow it must be ensured that the subcontracting of Software is done as proposed by the RB regulations as described in

- CD 02702 Outsourcing of Work
- CD 8006 Bosch Procurement and Supplier Management
- CD 04505 Open Source Software

[Process Step: Describe in detail the scope of work to be subcontracted (generic)]

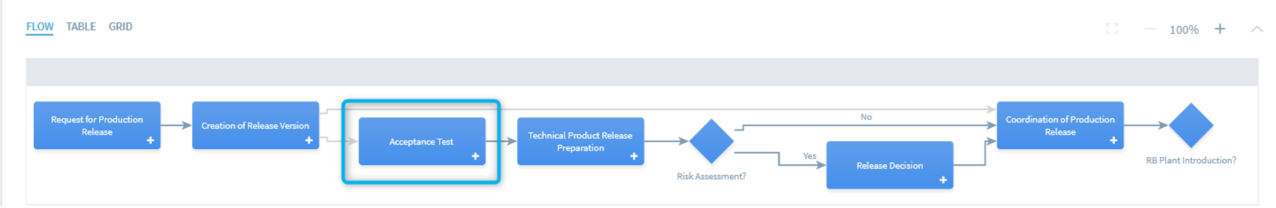
[Task: Agree on suitable suppliers for tendering with purchasing department (generic)]

[Process: During Development]

[Task: Conduct regular technical and schedule meetings (generic)]

[Workflow and Activities : Product Delivery]

Product Delivery

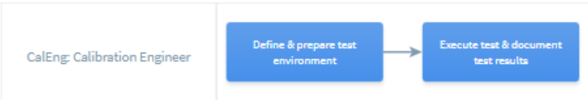


Task: Perform acceptance test (generic)

Process > Workflows and Activities > Product Delivery

Acceptance Test

FLOW TABLE GRID



DESCRIPTION

IN GENERAL

The Purpose of the Workflow *Acceptance test* is to validate the complete systemic relationships in the series version (series hex file) with the production release steps in the target system (vehicle).

Tailoring

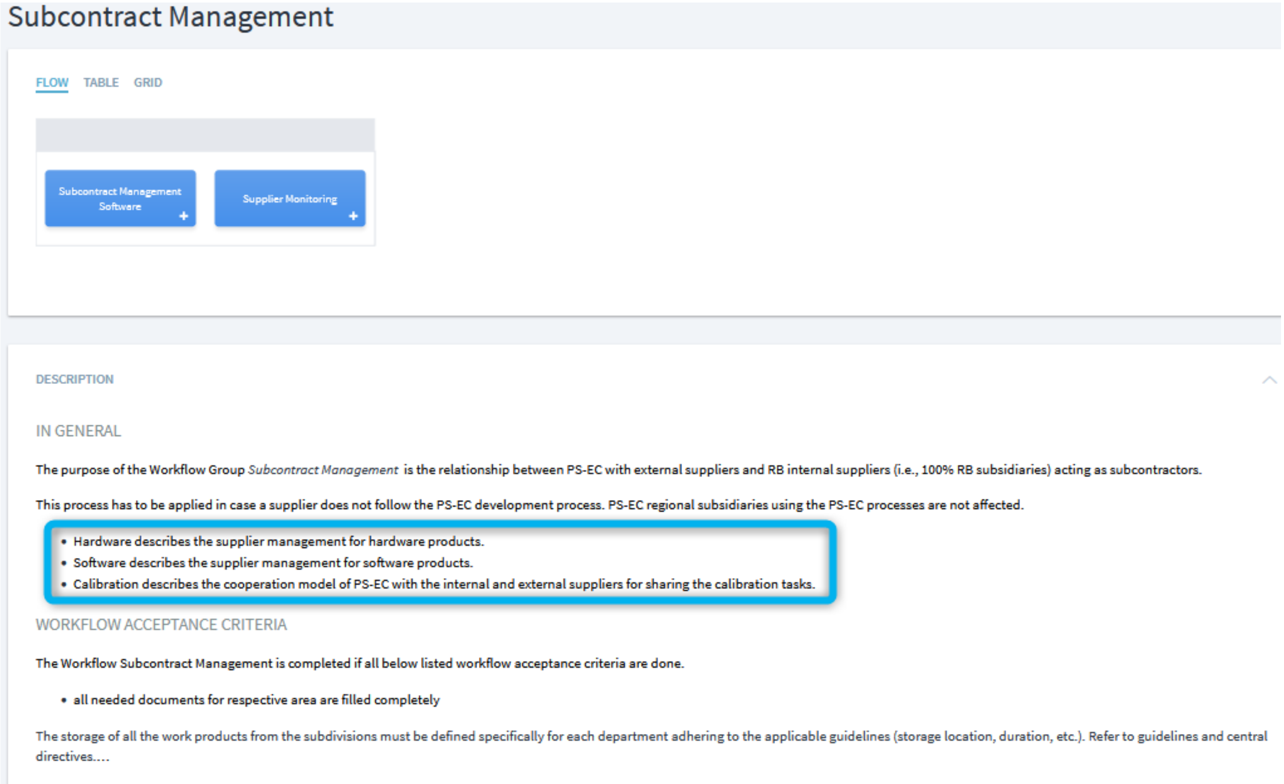
- The test execution for acceptance test (AT) is to be scaled and documented, based on the changes in case of data variants derived from the series SW. This includes the possibility to perform the AT exclusively for specific data sets. The selection and the decision criteria have to be documented in the project.
- Execute the test based on the project agreement, the responsibilities and the resulting works splits. Responsible for the definition of execution is the PjM-ECU.
- The agreements or assumptions have to be documented in writing by all parties.
- The acceptance test can also be performed outside of the PS-EC/CE business division.
- If there is no vehicle available to perform the AT at Bosch, the customer is responsible for the execution of the vehicle tests. In this case, an AT or a BFT at HiL has to be performed with the release version. This has to be documented in the series release document.
- The AT does not need to be performed if the final DST (PVER+data), which is the input product for the creation of the series hex files (Entry for the process "Creation of Release Version), is checked with the OT. For dataset variants which are evaluated according to the OT tailoring (OT based on changes or formal review) the project must decide if it is necessary to perform an AT for the final DST (PVER+Data).
- The AT does not need to be performed if the OT is carried out with the series hex files (Output of the process "Creation of Releaseversion").
- The AT does not need to be performed if Bosch is not responsible for the creation of the final DST (PVER+data).
- The AT cannot be performed if series SW is not a drivable program version (only startup block, customer boot block).

WORKFLOW ACCEPTANCE CRITERIA

The Workflow *Acceptance test* is completed if all below listed workflow acceptance criteria are fulfilled.

- Define and prepare test environment
- Execute test & document test results

Note – Changes are also specified in SW, HW and CAL Product development in Subcontract Management in all the 3 processes.



12 Important Links

Safety @PS-SC – Community

CoC Vehicle Safety

Functional Safety

CD 00214

13 Revision History

Issue	Date	Editor	Description of amendment
1.0	25/04/2013	Roshan Umredkar (RBEI/EPQ1)	Initial version
2.0	20/09/2013	Roshan Umredkar (RBEI/EPQ1)	Update based on PLIB V4.7 release. Sections updated – 5 to 6, 7.3, 9, 11
3.0	01/10/2015	Raghuram KN(RBEI/EPQ2)	Update based on PLIB V4.12 release.
4.0	30/11/2016	Raghuram KN(RBEI/EPQ2)	Update based on PLIB V5.0 release.
4.1	15/05/2017	Ahmed M Idris (RBEI/EPQ1)	Sections updated – 5 & 6 Updated 6-C.5.5 remarks Section 6 for CE/SW tests
4.2	06/11/2017	Ahmed M Idris (RBEI/EPQ1)	Updated sections : 5, 6 & 7
5.0	28.06.2019	Rajesh C Toms (RBEI/EMP1-EC)	Update according to ISO 26262- 2nd edition. All references to work products updated. Compatible with ProcLib V5.5
5.1	18.06.2020	Rajesh C Toms (RBEI/EMP1-EC)	Update for correction of links, Hints for tool classification, new WBT details added.
5.2	31.07.2020	Rajesh C Toms (RBEI/EMP1-EC)	Add RASIC in section 5.1, Update section 5.1 & 5.2 for quotation process changes.
5.3	16.12.2021 06.01.2022	Rajesh C Toms (MS/PJ-FS-PS) Steffan/Rigger (PS-EC/EHT2)	Update links, pictures, WBT details, and other improvements based on user feedback. Update links to HW part; also in stages V7.
5.4	16.09.2022	Sugantha M (MS/PJ-FS-PS)	Update based on PLIB stages V7 release
5.5	03.05.2023	Arun Kumar Muthu (MS/PJ-FS-PS)	Update of links in HW (5-6.5.1) part and Stages V7 Update of Tests in part 4-7.5.2 and 6-10.5.1 Update of generic url for all ProcLib links

5.6	18.08.2023	Arun Kumar Muthu (MS/PJ-FS-PS) Kaletka Tobias (PS-EC/EHT2)	Update of contents in HW from 5-7.5.1 until 5-10.5.2
5.7	04.10.2024	Arun Kumar Muthu (MS/PJ-FS-PS)	Update of Responsible for TSR and TSC from SW SD to SysArch. Updated the Organisation's name from PS-EC to PS-SC.
5.8	14.03.2025	Arun Kumar Muthu (MS/PJ-FS-PS)	Update of broken links in pages 19, 20 related to SysQT. Replaced SPjM HW -> PjM HW & HW INT -> HW DEV ENG due to HW SPICE changes.